

MD. Arshad

Computational Systems Biologist | AI Research Engineer

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Portfolio: sulkysubject.github.io | GitHub: github.com/Sulkysubject37

PROFESSIONAL SUMMARY

Computational Systems Biologist and Research Engineer focused on **mechanistic interpretability** and **computational tooling** for biological systems. Experience includes implementing deterministic C++ kernels (**VECTORIA v1.3.0**), reliability-first generative pipelines (**resED**), and the **resLIK** R package for latent manifold governance. Work bridges theoretical biology with systems-level implementation (SIMD/AVX2, Compiler IRs, Memory Arenas).

TECHNICAL SKILLS

- **Languages:** C++17/20 (STL, Templates, PyBind11), Python (NumPy/PyTorch Internals), Swift, R, Assembly (ARM64 NEON, x86_64 AVX2, SIMD).
- **Systems Engineering:** SIMD Optimization, Arena/Pool Memory Management, Compiler Concepts (IR/Lowering), Deterministic Computing.
- **Computational Biology:** De Novo Assembly (De Bruijn Graphs), GRN Inference (Causal/Temporal), Structural Biology (PDB/Ramachandran analysis).
- **Tools & DevOps:** CMake, Docker, Git, GitHub Actions (CI/CD), LLVM (basics).

ENGINEERING & RESEARCH EXPERIENCE

Research Engineer (Open Source)

Jan 2024 – Present

Remote / Independent

- **VECTORIA (v1.3.0-stable):** Designed a deterministic, inspectable computational kernel framework (C++17, SIMD, Assembly) enforcing bit-exact 'Reference-as-Truth' semantics.
 - Implemented a full **Semantic Inference Stack** including Transformer Encoder blocks using purely composed graph expansions without fused kernels.
 - Wrote validated **SIMD assembly kernels** (ARM64 NEON, x86_64 AVX2) with 100% pass rates on architecture-aware CI.
 - Designed a frozen Graph IR for auditable lowering to CoreML/MIL.
- **Jinxemblem (De Novo Transcriptome Assembler):** Developed a C++20 assembler using compact **De Bruijn Graphs (2-bit encoding)**.
 - Implemented **Tarjan's SCC algorithm** and **Maximal Non-Branching Path (MNP)** decomposition to resolve complex graph cycles without combinatorial explosion.
 - Achieved >99% precision/recall on synthetic benchmarks, focusing on low-memory footprint.
- **RLCS & resED (Reliability Engineering):** Formalized the **Representation-Level Control Surfaces (RLCS)** paradigm to treat AI reliability as a managed system property.
 - Developed the **resLIK R Package (v0.1.2)**: A CRAN-ready reference implementation for latent manifold monitoring using ResLik, TCS, and Agreement sensors.
 - Built **resED**: A production-grade reliability-first generative pipeline. Achieved 99.6% clean data acceptance while preserving 100% noise rejection in safety-critical loops.

- Designed O(d) forward-only sensors to identify distribution shifts and modal conflicts without retraining.

Bioinformatics Researcher
Independent

Jan 2024 – Present

- **KORA (Kinetic Ordered Regulatory Analysis):** Engineered a **Spiking Neural Network (SNN)** framework for inferring Gene Regulatory Networks. Replaced static correlation methods with **Spike-Timing Dependent Plasticity (STDP)** rules to capture causal temporal precedence in transcriptomic data.
- **C-STDP (Causal Inference Pipeline):** Specialized the STDP approach for infectious disease modeling (COVID-19), identifying upstream regulators of the host immune response via derivative-based Z-score encoding.

SELECTED PROJECTS

- **DNA Data Storage (v1.2-delta):** High-performance research prototype encoding binary data into DNA. Implemented Hamming/Reed-Solomon ECC and 'Rejection-and-Remap' logic for biochemical constraints.
- **Bio-Regulatory Layer (v1.2.1):** Production-grade secure reference implementation for drug regulatory analysis. Features IAM (SQLite-backed), rate-limiting, and isolated execution directories for safe concurrency.
- **annotaR (v0.1.1):** R package for functional enrichment analysis (GO/KEGG). Optimized for CRAN standards with DOI-backed evidence and high-fidelity dotplot visualizations.
- **tangle (C++ Network Analysis):** Engineered a high-performance PPI library with a custom TUI. Implemented parallelized **Louvain Community Detection**, outperforming Python equivalents in efficiency.

EDUCATION

Master of Science in Bioinformatics
Jamia Millia Islamia, New Delhi

2025 – 2027 (Ongoing)

Bachelor of Science in Bioinformatics
Maulana Abul Kalam Azad University of Technology (MAKAUT), WB

2021 – 2025

- **CGPA:** 8.49
- *Relevant Coursework:* Structural Biology, Biostatistics, Algorithm Design, Genetics.

RESEARCH PHILOSOPHY

I prioritize **causal interpretability** and **explicit computational primitives**. My goal is to build scientific software that is verifiable, debuggable, and mathematically rigorous.